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Compressor Performance Map Prediction using Machine Learning Algorithms

A Multi-Map Training Approach

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ABSTRACT

Compressor maps are often used beyond the conditions where complete measured or simulated data are available. Applications such as off-design analysis, transient simulation, and control development require fuller map coverage, making the prediction of missing speed lines and operating boundary regions, such as surge and choke, necessary. This work presents a configurable machine learning framework for reconstructing missing compressor map regions by combining the visible portion of a target map with information learned from other compressor maps. The main idea is to treat map reconstruction as a configurable prediction problem, where the support maps, auxiliary coordinate definition, set of input features, prediction model type, and hyperparameters are selected according to the missing data scenario. This allows the method to adapt to different prediction tasks and identify the policy combination that gives the most accurate prediction for the target case. The solution distinguishes the corrected mass flow–pressure ratio (W_c–R_c) geometry reconstruction problem from the efficiency (η_c) surface reconstruction problem and proposes a separate method for each. For geometry prediction, the method constructs a baseline estimate from the visible map and trains residual models to correct the predicted W_c and R_c, using an auxiliary coordinate, β, to describe point location along a missing speed line or endpoint extension. The framework is evaluated on seven missing data cases, including missing intermediate and exterior speed lines, and partially or fully missing surge and choke regions. For the Acp_4001 case study, the selected configurations give RMSE values of 0.0037–0.0643 for R_c and 0.0015–0.0176 for W_c, corresponding to reductions of 45.8–89.9% and 30.0–81.3%, respectively, relative to the visible data baseline. The selected efficiency methods give η_c RMSE values between 0.0020 and 0.0105 and also reduce the baseline error in all seven cases. The results show that the same reconstruction framework can be adapted to different types of missing map data by changing the selected training maps, coordinates, features, and models.

NOMENCLATURE

N_c	corrected rotational speed
W_c	corrected compressor mass flow
R_c	compressor pressure ratio
η_c	compressor efficiency
β	auxiliary point position coordinate
ANN	artificial neural network
BPNN	backpropagation neural network
ExtraTrees	Extremely Randomized Trees
GPR	Gaussian process regression
MAPE	mean absolute percentage error
ML	machine learning
RBF	radial basis function
RMSE	root mean square error
SVR	support vector regression
z	subscript denoting a standardised quantity
vis	subscript denoting a visible quantity
rel	subscript denoting a relative quantity
target	subscript denoting a target quantity

1.0 Introduction

Compressors are essential components of gas turbines and significantly influence overall engine performance, efficiency, and stability. Their aerodynamic behaviour over a range of operating speeds is commonly represented by performance maps relating corrected mass flow rate, pressure ratio, and efficiency. These maps are typically characterised through rig testing or high-fidelity simulation, both of which are costly and time consuming; measured points cluster around the intended operating region, and data are particularly difficult to acquire near the operating limits. Because the resulting characteristics also embody sensitive design knowledge, complete maps are rarely released by manufacturers. In practice, maps are often used beyond the conditions where complete data are available. Off-design analysis, transient simulation, and control development all require fuller map coverage than testing typically provides, making the reconstruction of missing speed lines and boundary regions, such as surge and choke, an important research topic within the gas turbine community.

Traditional compressor map prediction methods rely on interpolation and extrapolation to generate new speed lines, using formulations such as partial least squares regression [1] and Bézier-curve constructions with optimised control points [2]. Although useful for interpolation, these methods can become less reliable near operating boundaries where the prediction depends on extrapolation. A range of alternative methods has therefore been proposed, broadly divided into physics-based and data-driven approaches, with the latter receiving increasing attention because they learn the required relationships directly from performance data.

Artificial neural networks (ANNs) are among the most widely studied data-driven methods. Several ANN architectures have been evaluated for axial compressor map prediction [3, 4]. Yu et al. [5] investigated backpropagation neural networks (BPNNs), while Fei et

al. [6] combined a BPNN with a Gaussian kernel function to improve prediction with limited training data. Other machine learning algorithms have also been investigated. Ying et al. [7] reported that support vector regression (SVR) outperformed three ANN models in both interpolation and extrapolation, while Zhong et al. [8] found that SVR outperformed BPNNs and cubic spline interpolation. Recent compressor map prediction studies have developed several related directions: sequence-aware learning, physics-based parameterisation, transfer learning, direct use of map shape features, and physics-informed turbomachinery surrogates. Sequence-based work treats the speed line as an ordered sequence rather than a cloud of independent points; gated recurrent units have been shown to track the trend of a characteristic more faithfully than pointwise regressors [9], and surrogate-assisted hyperparameter optimisation of recurrent networks, with sequence-aware preprocessing of the map data, has been applied specifically to enhance interpolation and extrapolation of compressor performance maps [10]. In a related physics-based alternative, each speed line is compressed into a small set of interpretable superellipse parameters, such as surge point, choke point, and curvature, and the prediction is performed in that parameter space [11]. Data scarcity has also been addressed through transfer learning, where hybrid schemes combining recurrent networks with turbomachinery similarity laws have been shown to outperform stand-alone learners when only fragmentary operational records of the target machine exist [12]. Other approaches inject the geometry of the map directly into the model by fusing extracted shape features of the characteristics into an adaptively updatable performance-curve database [13]. Finally, physics-informed and CFD-trained machine learning for turbomachinery is maturing at industrial scale, from hybrid digital-twin frameworks [14] to deep learning surrogates of complete multi-stage axial compressors capable of reusing pre-trained models on data-poor applications [15]. Although these studies demonstrate the potential of machine learning for compressor map prediction, they generally develop and evaluate their models using data from a single compressor map.

In our previous work [16], four ANN architectures, SVR, and Gaussian process regression (GPR) were evaluated for single map prediction. That earlier study also introduced an auxiliary coordinate describing point position within the compressor map. The best performing ANN achieved a mean absolute percentage error (MAPE) of approximately 1%. However, training and evaluation were still performed using a single compressor map. The present study uses a different residual-regression framework within a multi-map reconstruction setting.

The present study builds on the previous work in two directions, each responding to a gap in the literature surveyed above. First, multiple radial compressor maps are used as support data during model training, rather than relying exclusively on the target map. This exposes the models to a wider range of speed line trends and provides additional training examples when the visible target data are limited, bringing the benefits pursued by transfer learning approaches [12] into the map reconstruction setting, and directly addressing the single map limitation shared by most prior studies. Second, the reconstruction problem is divided according to the missing data scenario because intermediate speed interpolation, exterior speed extrapolation, and endpoint completion impose different geometric constraints.

The main contribution is therefore not a single universal prediction model, but a configurable framework combining: (i) a baseline estimate constructed without using the withheld target points; (ii) an auxiliary coordinate, β, describing the relative position of each query point along a speed line or endpoint extension; (iii) residual learning using local geometric and map level features; and (iv) case-dependent use of target map and support map training data. Reconstruction is further divided into two stages: prediction of the map geometry, rep-

resented by W_c and R_c, followed by prediction of efficiency, η_c. This separation reflects the different roles of these variables: W_c and R_c define the speed line shape, whereas η_c forms a comparatively smooth surface over the compressor map.

2.0 Problem Definition and Data

2.1 Prediction tasks

In this work, we evaluate the framework using seven controlled missing data cases:

1. *Missing intermediate speed line*: one speed line between two visible speeds is removed.
2. *Missing lower exterior speed line*: the lowest speed line is removed.
3. *Missing upper exterior speed line*: the highest speed line is removed.
4. *Individual speed choke region*: the choke region of a single speed line is removed.
5. *Individual speed surge region*: the surge region of a single speed line is removed.
6. *Missing full choke region*: the choke region of all speed lines is removed.
7. *Missing full surge region*: the surge region of all speed lines is removed.

For each case, the removed points are completely hidden, and only the remaining points are used to construct the baseline predictions and input features. The visible target map provides the first source of information about the missing region. Support maps are then evaluated as an additional source of training examples, with their usefulness assessed separately for each missing data case.

2.2 Compressor map collection

The compressor maps used in this study were obtained from the GasTurb component map collection [17]. The collection documents compressor and turbine maps derived from open literature. The map labels used here, such as Acp_4001, are therefore file identifiers from the collection rather than standard compressor names.

The Acp_4001 radial compressor map is used as the target map and is shown in Figure 1, which illustrates the speed line, surge region, and choke region terminology used throughout the paper. We also use seven radial compressor maps as candidate support maps. Table 1 lists the maps and their characteristics. The full collection contains 1300 points, including 1160 points from the support maps. The map parameters are N_c, W_c, R_c, and η_c.

Because the compressors are of different sizes and operating ranges, the maps have different W_c and R_c ranges. To allow the model to learn corrections less dependent on scale, we normalise each map separately instead of normalising the full dataset using a single mean and standard deviation. For a variable x, the z-score standardised value is

$$x_z = \frac{x - \mu_{x,\text{vis}}}{\sigma_{x,\text{vis}}}, \qquad \ldots(1)$$

where x_z is the z-score standardised value of x, and $\mu_{x,\text{vis}}$ and $\sigma_{x,\text{vis}}$ are the mean and standard deviation of the visible points for that map and training case. Throughout this paper, the subscript z denotes a standardised quantity.

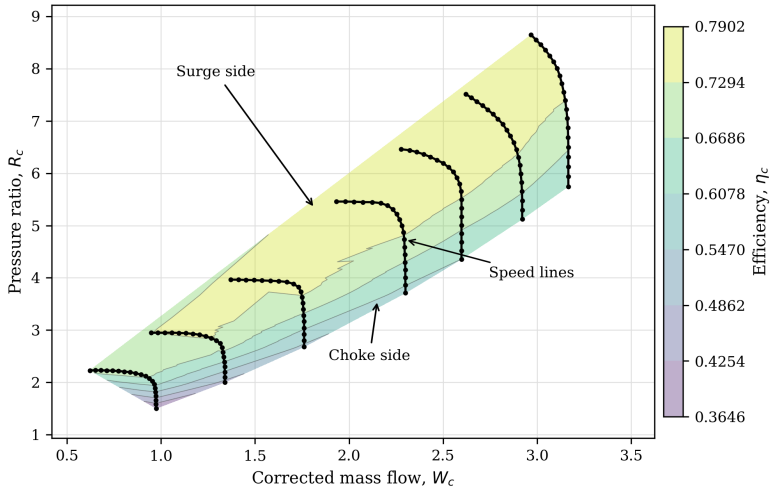


Figure 1. Representative target compressor map from the open-literature GasTurb map collection. The curves are speed lines in the W_c–R_c plane, and the shaded contours show compressor efficiency

Table 1
Target and candidate support compressor maps used in this study

Map	Reference	Speed lines	Points/line	Total points
Acp_4001 (target)	AGARD CP 400 [18]	7	20	140
ASME-94-GT-73	ASME 94-GT-073 [22]	6	20	120
Pep82r15	AGARD PEP 82 [24]	7	20	140
Rad1kg	NASA TM X-3552 [23]	8	20	160
RadComp10-AIAA-79-1159	AIAA 79-1159 [21]	9	20	180
RadComp9-AIAA-79-1159	AIAA 79-1159 [21]	8	20	160
T100rad	ASME 89-GT-112 [20]	6	20	120
T800_c	ASME 94-GT-202 [19]	14	20	280

3.0 Geometry Reconstruction of W_c and R_c

In this section, we explain the baseline construction, auxiliary β coordinate, and input features used to predict the W_c and R_c residuals. The shared training policies and model selection procedure are described later in Section 5.

3.1 Visible data baseline

For each missing data case, we first calculate a baseline prediction using linear interpolation or extrapolation from the visible map data. The machine learning model then predicts a correction to this baseline, rather than generating the missing speed line directly from the input features. This gives the model an initial physically plausible line location, and limits the learning task to correcting the remaining shape error. For a missing full speed line, all visible speed lines are first expressed on a common 20-point coordinate. The baseline is then obtained by

interpolating or extrapolating W_c and R_c with respect to N_c at each coordinate value:

$$\widehat{\mathbf{g}}_0(\beta) = \left[\begin{array}{cc} \widehat{W}_{c,0}(\beta) & \widehat{R}_{c,0}(\beta) \end{array} \right], \quad \ldots(2)$$

where $\widehat{\mathbf{g}}_0$ is the baseline geometry vector, the hat denotes an estimated quantity, and the subscript 0 identifies the baseline before applying the machine learning correction.

For choke and surge completion, the remaining part of the same speed line is used to estimate its missing region. Specifically, we calculate the local direction from the two visible points closest to the missing region, then extend the line by the required number of missing endpoint points to form the baseline. In the present test cases, this number is five points per affected speed line. This construction is identical for both individual speed and full region removal.

We define the machine learning outputs as the standardised residuals between the measured values and the baseline:

$$\Delta W_{c,z} = \frac{W_c - \widehat{W}_{c,0}}{\sigma_{W,\text{vis}}}, \quad \ldots(3)$$

$$\Delta R_{c,z} = \frac{R_c - \widehat{R}_{c,0}}{\sigma_{R,\text{vis}}}. \quad \ldots(4)$$

Here, $\Delta W_{c,z}$ and $\Delta R_{c,z}$ denote the standardised W_c and R_c residuals, respectively. The quantities $\sigma_{W,\text{vis}}$ and $\sigma_{R,\text{vis}}$ are the corresponding standard deviations calculated from the visible points. After models yield the prediction, the standardisation is reversed to recover the residuals in physical units. These residuals are then added to the baseline values to obtain the final predicted W_c and R_c.

3.2 Auxiliary beta coordinate

The relationship between R_c and W_c along a speed line can become multi-valued in the extremities of the map. Near surge, a speed line can become nearly horizontal, while near choke, the line can become nearly vertical. We therefore add an auxiliary coordinate, β, to identify the relative position of each point along the predicted line. The β coordinate used here should not be confused with conventional beta lines or beta parabolas used in some compressor map representations; here, it is only a point-position coordinate calculated from the visible data baseline and therefore remains producible when the target points are missing.

We propose several possible methods for defining β, illustrated in Figure 2. For missing full speed line cases, the candidate definitions order the baseline points from the lowest R_c endpoint to the highest R_c endpoint and scale β from zero to one. Endpoint completion uses a separate local definition because only the missing endpoint segment is predicted.

Figure 2(a), uniform point order: This definition assigns equally spaced β values according to the ordered point index. It assumes that the speed lines used for training and prediction have first been represented using a common number of points:

$$\beta_i^{\text{uniform}} = \frac{i-1}{n-1}, \qquad i = 1, \ldots, n. \quad \ldots(5)$$

Here, n is the number of points on the baseline and i is the point index, ordered from the lowest to the highest R_c endpoint.

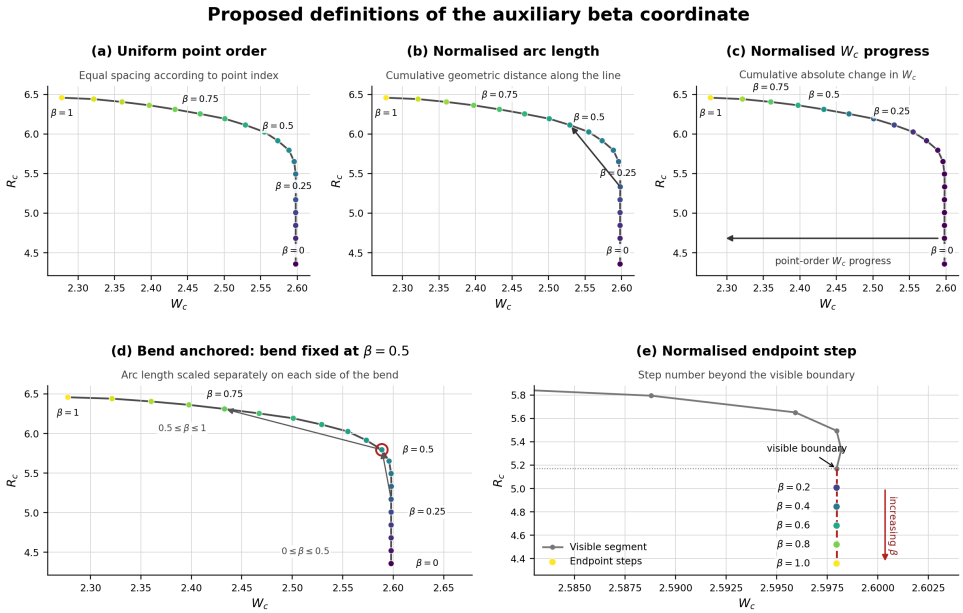


Figure 2. Candidate definitions of the auxiliary β coordinate illustrated using the 95% speed line of Acp_4001: (a) uniform point order, (b) arc length, (c) mass flow progress, (d) bend anchored arc length, and (e) endpoint step. In the full speed line definitions, β is scaled from the lowest to the highest R_c endpoint. For endpoint completion, β increases locally with distance beyond the visible boundary; the choke region is shown as an example

Figure 2(b), arc length: This definition uses the cumulative geometric distance along the baseline in the normalised W_c–R_c plane:

$$\beta_i^{\text{arc}} = \frac{\sum_{j=2}^{i} \left\| \widehat{\mathbf{g}}_{0,z,j} - \widehat{\mathbf{g}}_{0,z,j-1} \right\|_2}{\sum_{j=2}^{n} \left\| \widehat{\mathbf{g}}_{0,z,j} - \widehat{\mathbf{g}}_{0,z,j-1} \right\|_2}. \qquad \ldots(6)$$

Here, $\widehat{\mathbf{g}}_{0,z,j} = [\widehat{W}_{c,0,z,j}, \widehat{R}_{c,0,z,j}]$ is point j on the standardised baseline, and $\|\cdot\|_2$ denotes Euclidean distance.

Figure 2(c), mass flow progress: This definition uses the cumulative absolute change in W_c following the same point order:

$$\beta_i^{\text{flow}} = \frac{\sum_{j=2}^{i} \left| \widehat{W}_{c,0,j} - \widehat{W}_{c,0,j-1} \right|}{\sum_{j=2}^{n} \left| \widehat{W}_{c,0,j} - \widehat{W}_{c,0,j-1} \right|}. \qquad \ldots(7)$$

Figure 2(d), bend anchored arc length: Let $s_i = \beta_i^{\text{arc}}$, and let κ_i be the estimated baseline curvature at point i. The same endpoint ordering is used as in the other full speed line definitions, so s_1 corresponds to the lowest R_c endpoint and s_n to the highest R_c endpoint. The point of

greatest curvature is $k = \arg\max_i \kappa_i$. The arc length on either side of this point is rescaled as

$$\beta_i^{\text{bend}} = \begin{cases} \frac{1}{2}\frac{s_i - s_1}{s_k - s_1}, & i \le k, \\ \frac{1}{2} + \frac{1}{2}\frac{s_i - s_k}{s_n - s_k}, & i > k. \end{cases} \quad \ldots(8)$$

The bend is therefore assigned $\beta = 0.5$, while the two line segments are mapped to $0 \le \beta \le 0.5$ and $0.5 \le \beta \le 1$, respectively.

Figure 2(e), endpoint step: For endpoint completion, β is defined as the normalised extrapolation step:

$$\beta_j^{\text{end}} = \frac{j}{m}, \qquad j = 1, \ldots, m, \quad \ldots(9)$$

where j is the step beyond the visible boundary and $m = 5$ is the number of hidden points. This endpoint coordinate is local: β increases from the visible boundary towards the farthest predicted point, irrespective of whether the missing region is at surge or choke.

Across these definitions, β is always calculated from the visible data baseline rather than the hidden target points. The full speed line definitions also assume a common point count and consistent endpoint ordering across the training and prediction examples.

Different definitions of β may be more suitable for different missing data cases. Therefore, all proposed definitions are evaluated separately for each case, allowing the most suitable β coordinate to be selected together with the model type, feature set, and support map policy. The selected definitions for the Acp_4001 case study are reported later with the final geometry configurations.

3.3 Input features

Feature construction depends on the missing data case. For a missing full speed line, we test the *shape* and *full* feature sets. The *shape* feature set contains the target speed location, auxiliary β terms, normalised baseline point, and local baseline geometry. In simplified form, the input vector of a query point is

$$\mathbf{x}_{\text{shape}} = [N_{c,\text{rel}}, N_{c,z}, \beta, \beta^2, \beta(1-\beta), \widehat{W}_{c,0,z}, \widehat{R}_{c,0,z}, W_0', R_0', s_0, \kappa_0]. \quad \ldots(10)$$

The relative target speed is calculated as $N_{c,\text{rel}} = (N_{c,\text{target}} - N_{c,\text{min,vis}})/(N_{c,\text{max,vis}} - N_{c,\text{min,vis}})$, while $N_{c,z}$ is obtained using Eq. (1). The prime denotes differentiation along the common point coordinate used to construct the baseline, $s_0 = R_0'/W_0'$ is the local slope, and κ_0 is the local curvature. For choke and surge completion, we test the *direction* and *full* feature sets. The *direction* set uses the endpoint coordinate, baseline point, visible boundary point, and local direction of the visible speed line instead of the full line derivative features. In both types of task, the *full* set adds four visible map descriptors to the applicable base set: *shape* for a missing full speed line, or *direction* for endpoint completion. The *shape* and *direction* sets therefore contain 11 and 10 inputs, respectively, while their *full* versions contain 15 and 14 inputs. Figure 3 shows the geometric interpretation of the feature sets, and Table 2 describes the individual inputs.

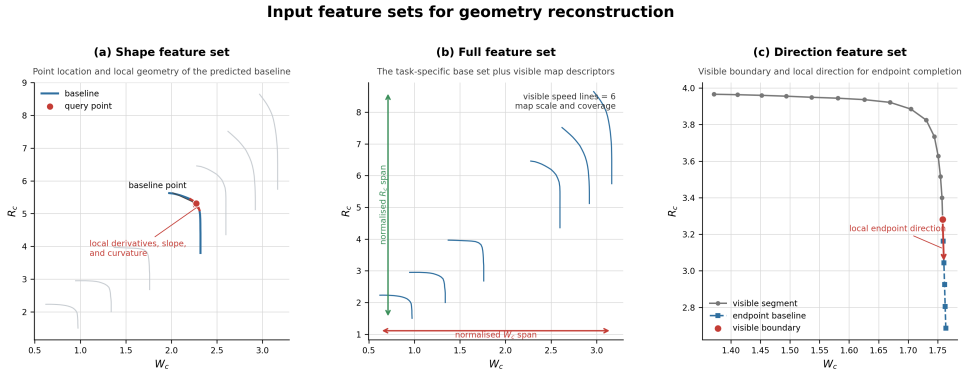


Figure 3. Construction of the geometry input feature sets using Acp_4001 examples. The shape set describes each baseline point and its local geometry. The full set augments the applicable task-specific feature set with descriptors calculated from the visible map. The direction set describes the visible boundary and local extension direction used for surge and choke completion

4.0 Efficiency Reconstruction

In this section, we reconstruct compressor efficiency after predicting the map geometry. Unlike W_c and R_c, efficiency forms a surface across the compressor map. We evaluate two approaches with different levels of complexity: a single common surface that provides a simple method for all missing data cases, and casewise residual models that require separate configurations but can provide higher prediction accuracy.

For each incomplete map case, the first approach fits a radial basis function (RBF) surface,

$$\widehat{\eta}_c = f_{\mathrm{RBF}}(N_c, \beta), \qquad \ldots(11)$$

where f_{RBF} is the fitted RBF surface and $\widehat{\eta}_c$ is the predicted compressor efficiency. The surface is fitted using the visible efficiency values of the target map. For this surface, β is a uniform point-order coordinate between zero and one. We use the same formulation and hyperparameters for all seven missing data cases. This provides a simple method that does not require a separately selected model for every case. In the individual speed region tests, the surface retains values from the corresponding regions of neighbouring target speed lines. In the full region tests, it must extrapolate towards a boundary that is absent from the visible target map.

The second approach follows the same baseline and residual learning procedure used for W_c and R_c. We first calculate an efficiency baseline from the visible data, then use a prediction model to estimate $\eta_c - \widehat{\eta}_{c,0}$. The candidate models and configuration selection procedure are described in Section 5. This method requires a separate configuration for each missing data case, but allows the model to learn behaviour that is not captured by one common surface, particularly during exterior extrapolation.

For the casewise efficiency residual models, the input features follow the same structure as the geometry residual models, but the output is the efficiency residual, $\eta_c - \widehat{\eta}_{c,0}$. The input vector includes the target speed, auxiliary β coordinate, baseline location, and the applicable local geometry or endpoint direction features. When the *full* feature set is used, the same visible map descriptors are added. The common RBF surface uses a simpler input space consisting only of N_c and a uniform point order β coordinate.

Finally, the predicted efficiency values are paired with the predicted W_c–R_c geometry using

Table 2
Input features used for W_c and R_c residual prediction

Feature	Construction	Information provided to the model	Included in
$N_{c,\mathrm{rel}}$	Target N_c relative to the minimum and maximum visible speeds	Position of the target speed within, below, or above the visible speed range	Shape, direction
$N_{c,z}$	Target N_c standardised using the visible speed mean and standard deviation	Target speed location relative to the distribution of visible speeds	Shape, direction
$\beta, \beta^2, \beta(1-\beta)$	Auxiliary point coordinate and nonlinear transformations; endpoint cases use β and β^2	Position along the predicted line and a nonlinear representation of that position	Shape, direction
$\widehat{W}_{c,0,z}, \widehat{R}_{c,0,z}$	Baseline point standardised using the visible map statistics	Approximate physical location of the point before residual correction	Shape, direction
W_0', R_0'	First derivatives of the baseline with respect to its common point coordinate	Local change of W_c and R_c along the baseline	Shape
s_0, κ_0	Local baseline slope and curvature	Local orientation and bending of the predicted speed line	Shape
Boundary point	Last visible W_c and R_c values next to the missing endpoint region, after standardisation	Location from which surge or choke completion begins	Direction
Direction vector	Difference between the two visible points closest to the missing region, scaled using the visible map standard deviations	Local direction in which the endpoint baseline is extended	Direction
Map descriptors	Normalised W_c and R_c spans, visible N_c span (maximum minus minimum), and number of visible speed lines	Scale, operating range, and speed line coverage of the visible map	Full augmentation

the same speed, case, and query point ordering. Thus, each reconstructed map point contains a predicted W_c, R_c, and η_c, although the final predicted geometry is not used as a direct input to the efficiency model.

5.0 Training and Configuration Selection

The geometry residual models defined in Section 3 and the casewise efficiency residual models are configured separately for each missing data case. The configurable choices include the support map policy, auxiliary coordinate definition, feature set, prediction model type, and hyperparameters. The common RBF efficiency surface is treated separately because it uses one fixed surface formulation across all cases.

5.1 Support map policies

For each test scenario, we compare model performance when using the target map only, one individual support map, the three or five nearest support maps by descriptor distance, or all support maps for training. To identify the nearest maps by descriptor distance, we construct a descriptor vector for each map using its number of speed lines, points per line, speed range, normalised flow and pressure ratio spans, and mean slope and curvature. We standardise the descriptors and rank the support maps according to their Euclidean distance from the target map descriptor. This is a descriptor based map ranking rather than a direct curve shape distance. For full choke and surge removal, the target descriptor is calculated using only the retained region, so the hidden boundary does not influence map selection.

To create training samples, we artificially hide known regions of the available maps. For example, we hide an interior speed line of a complete support map and calculate its baseline using the remaining lines. Since the hidden line is known during training, its difference from the baseline provides the required residual output. For upper exterior extrapolation, we create additional pseudo exterior tasks by hiding a high speed line and retaining only the lower speed lines. This gives the model training examples with the same extrapolation direction as the target case. For individual speed region completion, residuals can also be learned from the corresponding visible regions of the other target speed lines. For full region completion, all target labels on the corresponding boundary are excluded and residual learning uses support maps only.

Accordingly, Table 3 uses three training strategy labels. *Standard residual learning* refers to residuals obtained from the usual artificial masks of visible target or support maps. *Pseudo exterior residual learning* refers to the additional exterior-like masks created for upper exterior extrapolation. *Support map only full region residual learning* refers to full boundary removal cases in which target labels from the removed boundary are excluded from training.

5.2 Models, hyperparameters, and selection metrics

The tested residual prediction models are linear regression, Ridge regression, Extremely Randomized Trees (ExtraTrees), Random Forest, Gradient Boosting, and Gaussian process regression (GPR). This set provides both simple linear references and nonlinear ensemble/kernel models, allowing the comparison to distinguish whether the residual is adequately described by a low-order trend or requires a more flexible regression model. For geometry prediction, each model predicts both $\Delta W_{c,z}$ and $\Delta R_{c,z}$. For casewise efficiency prediction, the model predicts $\eta_c - \widehat{\eta}_{c,0}$.

We tune the hyperparameters separately for each missing data case. The tuned parameters include the number of trees, feature subsampling, minimum leaf size, tree depth, learning rate, regularisation strength, and kernel choice where applicable. Different support map policies, β definitions, and feature sets are also tested.

For geometry prediction, RMSE is calculated separately for every withheld speed line or endpoint segment and then averaged across the evaluated speed lines. We denote these averages by $\overline{\mathrm{RMSE}}_{R_c}$ and $\overline{\mathrm{RMSE}}_{W_c}$. Because R_c and W_c are different physical quantities, their RMSE values are reported separately in the results. A scalar selection score is used only to rank candidate configurations during the configuration search:

$$\mathcal{S}_{\mathrm{geom}} = \frac{1}{2}\left(\overline{\mathrm{RMSE}}_{R_c} + \overline{\mathrm{RMSE}}_{W_c}\right). \qquad \ldots(12)$$

Table 3
Selected geometry configurations for the Acp_4001 case study

Case	Training strategy	Support maps	Beta/coordinate	Feature set	Model
Intermediate speed	Standard residual learning	All support maps	Bend anchored arc length	Full	ExtraTrees (140 trees)
Lower exterior speed	Standard residual learning	Three nearest maps by descriptor distance	Uniform beta	Shape	ExtraTrees (160 trees)
Upper exterior speed	Pseudo exterior residual learning	Three nearest maps by descriptor distance	Mass flow progress beta	Shape	ExtraTrees (140 trees)
Individual speed choke region	Standard residual learning	Three nearest maps by descriptor distance	Normalised endpoint step	Full	ExtraTrees (140 trees)
Individual speed surge region	Standard residual learning	Target map only	Normalised endpoint step	Full	Linear regression
Full choke region	Support map only full region residual learning	All support maps	Normalised endpoint step	Direction	Linear regression
Full surge region	Support map only full region residual learning	Single selected support map (Rad1kg)	Normalised endpoint step	Full	ExtraTrees (140 trees)

Here, the two raw RMSE terms are weighted equally for ranking purposes, and a lower $\mathcal{S}_{\text{geom}}$ is used only to rank candidate configurations. The score in Eq. (12) is not interpreted as a physical error; the individual raw RMSE values remain the reported performance metrics. For efficiency prediction, configurations are selected using the mean raw η_c RMSE over the corresponding withheld speed line groups.

6.0 Results and Discussion

In this section, we evaluate the geometry and efficiency predictions using numerical errors and visual comparisons with the hidden target points.

6.1 Geometry prediction

For each case, we evaluate the candidate support map policies, β definitions, feature sets, models, and hyperparameters, and report the configuration with the lowest geometry selection score from Eq. (12). Table 3 lists the selected geometry configurations for Acp_4001. We use this case study to demonstrate how the framework can be configured for different prediction problems. These configuration options allow the prediction method to be customised according to the characteristics and missing data case of any new compressor map.

Figure 4 compares the selected predictions with the baselines and hidden target points. The intermediate speed subfigure overlays the five interior speed predictions, each obtained by removing one speed line at a time. Similarly, the individual speed choke and surge subfigures overlay seven predictions generated by removing the region from one speed line at a time while retaining it on the other lines. These subfigures are therefore composite visualisations of separate validation runs. In contrast, each full region subfigure represents one incomplete map in which all 35 points are removed simultaneously. For the lower exterior speed, the model corrects the large position error of the linear baseline. The upper exterior prediction also follows the hidden line more closely than the baseline.

Figure 5 shows one enlarged representative example from each missing data case to make the relative positions of the baseline, prediction, and hidden target points easier to inspect. When several speed lines or endpoint segments are evaluated within a case, the example shown is the one with the median geometry selection score among the selected predictions. The lower and upper exterior cases each contain one removed speed line and are therefore shown directly. The selected corrected speed is indicated in each panel title.

Table 4 lists the RMSE calculated separately for each withheld speed line or endpoint segment and then averaged within each case. For intermediate speed prediction, the table averages the five removable interior speed lines. Each exterior case contains one removed speed line. For the individual and full region cases, the RMSE is calculated for the endpoint segment

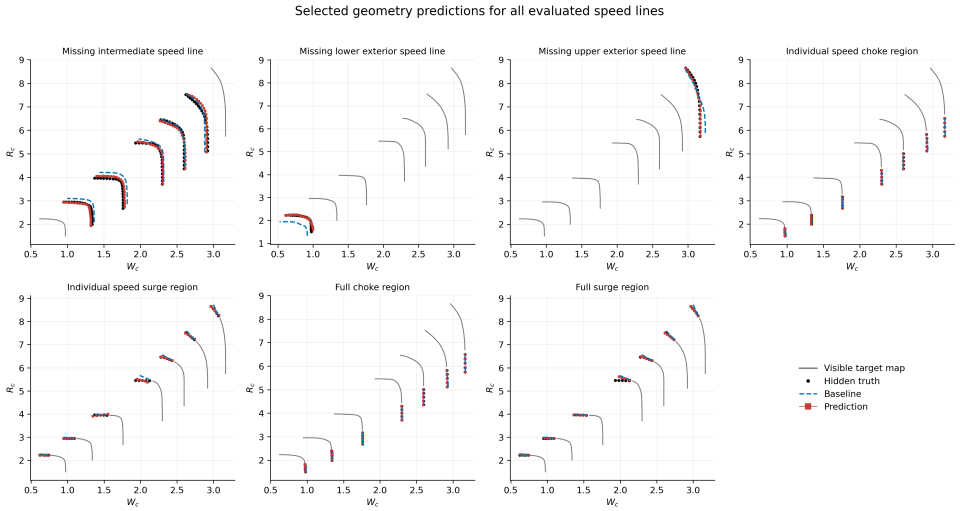


Figure 4. Selected W_c–R_c predictions for the seven missing data cases. The intermediate speed and individual speed region subfigures combine predictions from separate masking runs, whereas the full region subfigures show simultaneous removal from all speed lines

Table 4
Geometry prediction errors relative to the visible data baseline

	R_c RMSE			W_c RMSE		
Case	Baseline	Selected	Reduction (%)	Baseline	Selected	Reduction (%)
Intermediate speed	0.1689	0.0643	61.9	0.0344	0.0176	48.9
Lower exterior speed	0.2975	0.0300	89.9	0.0640	0.0120	81.3
Upper exterior speed	0.1123	0.0521	53.6	0.0429	0.0086	80.0
Individual speed choke region	0.0093	0.0037	59.5	0.0037	0.0015	60.0
Individual speed surge region	0.0515	0.0198	61.5	0.0121	0.0072	40.5
Full choke region	0.0093	0.0050	45.8	0.0037	0.0017	54.9
Full surge region	0.0515	0.0229	55.5	0.0121	0.0085	30.0

of each speed line and then averaged over the seven speed lines. The largest improvement is obtained for lower exterior extrapolation, where R_c RMSE decreases from 0.2975 to 0.0300 and W_c RMSE decreases from 0.0640 to 0.0120. Full surge region completion gives the smallest relative improvement in W_c, although its error is still reduced by 30.0%.

The variation across cases indicates that the difficulty of a missing map problem depends not only on the number of removed points, but also on whether the missing region is locally constrained by visible target data. Individual endpoint completion gives small absolute errors, which is consistent with the visible part of the same speed line anchoring the baseline. The exterior speed cases select support maps, and the upper exterior case additionally selects pseudo exterior training examples with the same extrapolation direction as the target task. The relatively smaller improvement in full surge W_c suggests that surge-side flow placement remains difficult when the complete boundary is unavailable.

The same baseline and selected geometry errors are compared visually in Figure 6.

Figure 7 compares the best hyperparameter setting of each model for every case using the

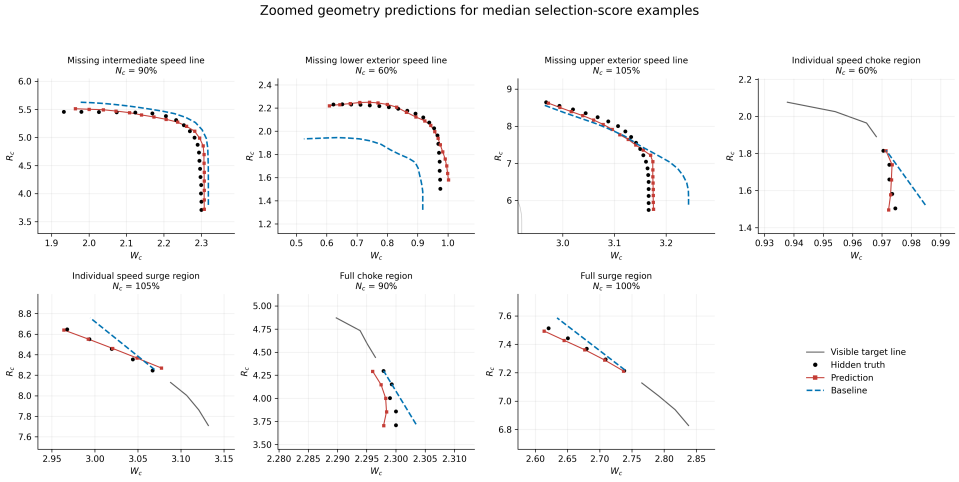


Figure 5. Zoomed W_c–R_c predictions for representative examples from the seven geometry cases. For cases with several evaluated speed lines or endpoint segments, the panel shows the median selection-score example; the selected corrected speed is shown in the panel title

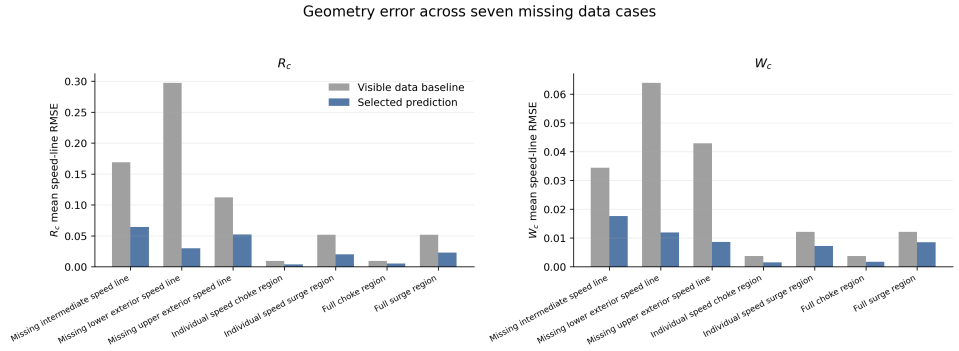


Figure 6. Comparison of baseline and selected mean geometry errors over speed line groups for the seven cases

same geometry selection score. ExtraTrees gives the lowest score for intermediate speed, both exterior speeds, individual speed choke completion, and the full surge region. Linear regression gives the lowest score for individual speed surge completion and the full choke region. This indicates that no single regression model is preferred for all missing data cases.

The selected configurations also show that support map use is case dependent. Intermediate speed interpolation uses all support maps, which provide multiple artificially hidden interior speed lines for training. Both exterior cases and individual speed choke completion use the three nearest maps by descriptor distance. The upper exterior case also uses pseudo exterior training, showing the importance of providing examples with the correct extrapolation direction. Individual speed surge completion uses linear regression and training data from the target map because the corresponding regions of the other target speed lines remain visible.

These results extend, rather than directly replace, the previous single-map study [16]. That work used the T-MATS high pressure compressor map, augmented each speed line through

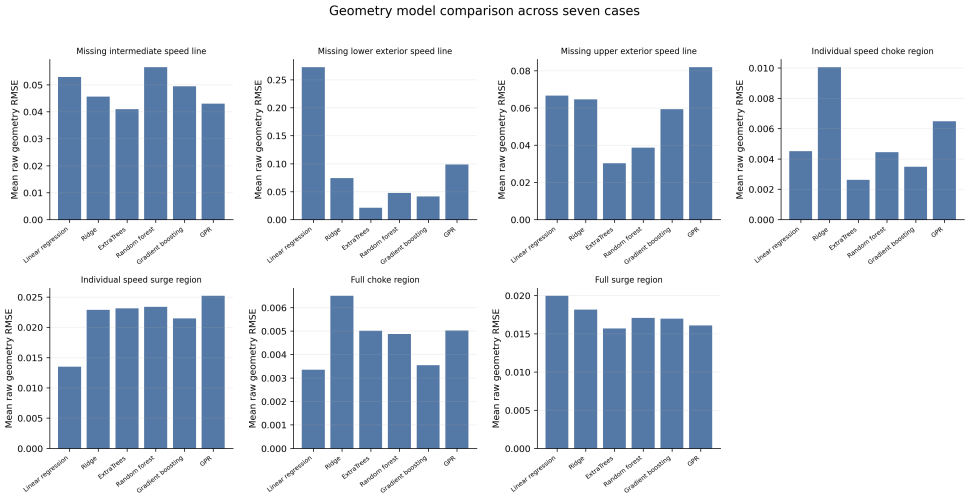


Figure 7. Geometry model comparison for the seven cases. Each bar shows the lowest geometry selection score obtained for that model type

polynomial interpolation, and predicted corrected mass flow from pressure ratio, corrected speed, and a beta-line coordinate, with the best models reaching approximately 1% MAPE. The present study addresses a different reconstruction setting: both W_c and R_c are predicted for missing speed lines and missing boundary regions, the target map is drawn from an open-literature map collection, and support maps are used when the visible target data are insufficient. The comparison therefore shows a change in scope more than a direct change in the same error metric. The framework moves from single-map corrected flow prediction to configurable multi-map reconstruction of compressor map geometry.

6.2 Efficiency prediction

Table 5 compares the visible data efficiency baseline, common RBF surface, and casewise residual method using the mean RMSE over speed line groups, and identifies the selected efficiency method for each case. The RBF surface gives the lowest error for intermediate speed and individual speed surge completion, with values of 0.0046 and 0.0025, respectively. The intermediate speed methods are nearly equivalent: when the errors are pooled over all hidden points instead, the RMSE is 0.00501 for the RBF surface and 0.00497 for the casewise method. The casewise residual method gives the lowest error for the remaining five cases, including full choke and surge completion with RMSE values of 0.0020 and 0.0040.

Figure 8 compares these efficiency errors for the visible baseline, the common RBF surface, and the selected casewise residual method.

The results show a trade off between simplicity and prediction accuracy. The RBF surface provides one efficiency formulation for all missing data cases and performs well when the missing efficiency values lie within a region still constrained by visible target map data. This explains its performance for the intermediate speed case, where the removed speed line lies between visible speed lines, and for individual speed surge completion, where neighbouring target speed lines still contain surge region efficiency values. Its accuracy deteriorates

Table 5
Mean efficiency RMSE over speed line groups and selected efficiency method for each case

Case	Visible baseline	RBF surface	Casewise residual	Lowest error prediction method
Intermediate speed	0.0087	0.0046	0.0049	RBF surface
Lower exterior speed	0.0185	0.0249	0.0105	Casewise residual
Upper exterior speed	0.0147	0.0134	0.0042	Casewise residual
Individual speed choke region	0.0106	0.0111	0.0023	Casewise residual
Individual speed surge region	0.0080	0.0025	0.0047	RBF surface
Full choke region	0.0106	0.0206	0.0020	Casewise residual
Full surge region	0.0080	0.0141	0.0040	Casewise residual

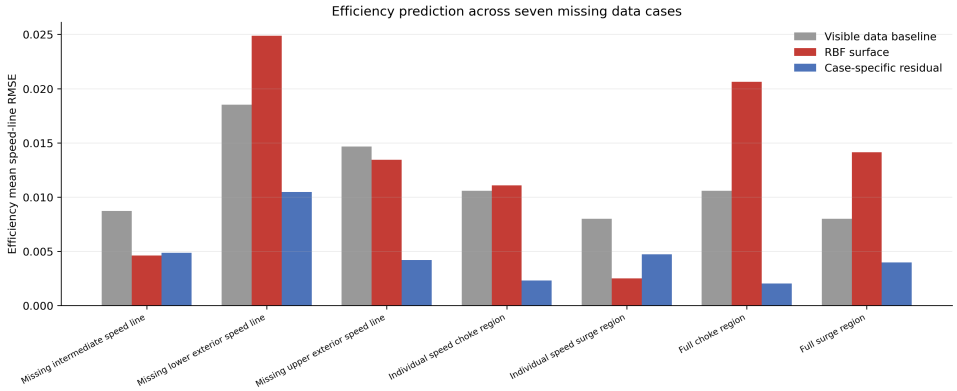


Figure 8. Mean efficiency RMSE over speed line groups for the visible data baseline, a single RBF surface formulation used across all cases, and the casewise residual method

when the problem requires stronger extrapolation, especially when the complete choke or surge boundary is removed. The casewise method requires a separate configuration but can use residual trends from the selected training maps when the common surface formulation is insufficient.

Figure 9 presents the same outputs as reconstructed efficiency contours obtained by combining the visible target data with the selected predictions. As in Figure 4, the intermediate and individual speed region subfigures combine predictions from separate masking runs. The full choke and surge subfigures each represent one simultaneous full region removal.

6.3 Limitations and future work

In this work, we evaluate the framework using one target compressor map. We select the final configurations after comparing them for Acp_4001. Thus, the reported results represent the selected performance for this case study and not an independent estimate of generalisation to an unseen map. For the full region cases, map ranking is calculated from visible geometry only; nevertheless, the final policy and model are selected using the withheld Acp_4001 values.

An automated configuration selection procedure was also examined, in which the support map policy, β definition, feature set, model, and hyperparameters were selected using validation cases generated from the available maps. However, for Acp_4001, this automated

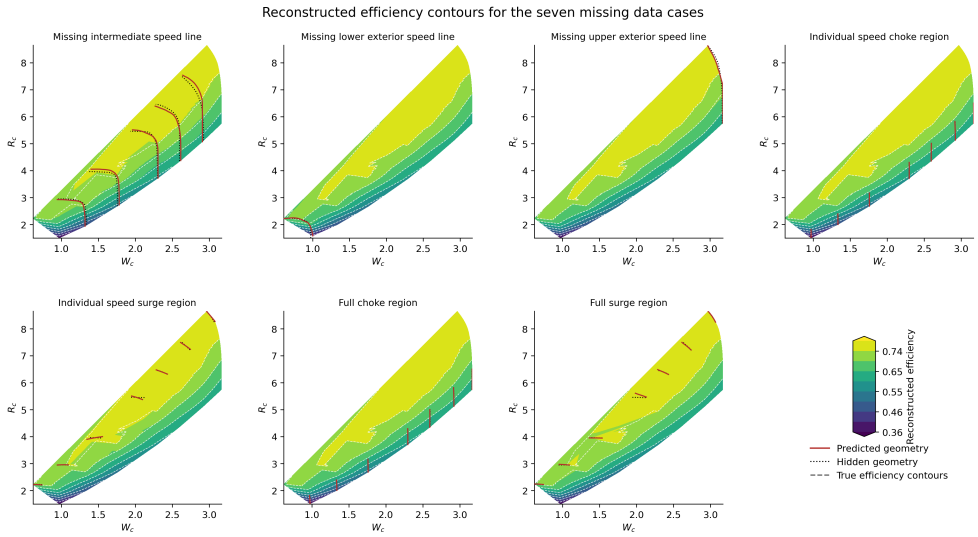


Figure 9. Reconstructed efficiency contours for the seven missing data cases

selection did not always recover the configuration that gave the lowest error when the withheld target map data were used for evaluation. It gave similar errors for the lower exterior case, where R_c RMSE changed from 0.0300 to 0.0315 and W_c RMSE from 0.0120 to 0.0133, but larger errors in some cases, especially upper exterior extrapolation, where the corresponding errors increased from 0.0521 to 0.2603 and from 0.0086 to 0.0267. Therefore, the results reported in this paper use the selected best performing configuration for the Acp_4001 case study. For a genuinely incomplete map, where the hidden region is unavailable for comparison, automated configuration selection remains useful as a practical guide, but it should be interpreted as a recommendation rather than a guarantee of the globally best configuration.

The present study also predicts the W_c–R_c geometry and efficiency surface separately. Although this separation is useful because the speed line geometry and efficiency contours have different behaviours, future work should verify the consistency between the two reconstructed quantities. For example, the predicted efficiency values should remain coherent with the reconstructed speed line locations and should not introduce unrealistic contour shapes.

All support maps are radial compressor maps with broadly similar speed line shapes. Therefore, the results cannot yet be generalised to arbitrary axial compressors, variable geometry maps, or maps with substantially different parameter ranges. The test cases are also created by removing points from regularly sampled 20-point speed lines. Fragmented or irregularly sampled data require further testing. Finally, future work should include physical consistency checks or constraints, particularly near surge and choke, where low pointwise RMSE may still produce speed lines with unrealistic slope, ordering, monotonicity, or boundary behaviour.

7.0 Conclusions

This work presented a configurable framework for reconstructing missing compressor map geometry and efficiency, combining the visible portion of a target map with support maps from other compressors. From the seven missing data cases evaluated on the Acp_4001 radial

compressor map, the main conclusions are:

1. Predicting residual corrections to a visible data baseline improves both full speed line reconstruction and endpoint completion. The selected configurations reduce R_c RMSE by 45.8–89.9% and W_c RMSE by 30.0–81.3% relative to the baseline, with the largest gains obtained for exterior speed extrapolation, where the linear baseline is weakest.
2. The auxiliary β coordinate provides point position information required where speed lines become multi-valued, without relying on a measured quantity. The suitable β definition depends on the missing data scenario, with different definitions selected for interpolation, exterior extrapolation, and endpoint completion.
3. The seven test cases require different configuration choices. Individual speed region completion and full region completion are not equivalent because they retain different amounts of target map information near the missing endpoint. Similarly, exterior speed prediction requires different treatment from intermediate speed interpolation.
4. Support maps are useful as a configurable source of training examples, but their role is case dependent. Several selected configurations use support maps, while the individual speed surge geometry case is best reconstructed using the visible target map only.
5. No single prediction model is preferred for all geometry cases. ExtraTrees gives the lowest error in five cases, while linear regression is selected for individual speed surge and full choke completion. This supports treating model selection as part of the framework rather than fixing one regressor in advance.
6. Efficiency prediction benefits from being treated separately from W_c–R_c geometry prediction. A common RBF surface is suitable when visible target map data still constrain the missing efficiency region, while casewise residual learning is more accurate for the remaining cases. The selected efficiency methods give η_c RMSE values between 0.0020 and 0.0105.
7. Support-validation-based automated configuration selection provides a practical recommendation procedure for genuinely incomplete maps, but it did not recover the lowest-error Acp_4001 configuration in this retrospective study. The reported paper results therefore represent the selected Acp_4001 case-study performance rather than a fully automatic guarantee for unseen maps.

Future work will address three main extensions: generalisation beyond a single target map and beyond radial machines, enforcement of physical consistency near surge and choke, and coherence between the separately reconstructed geometry and efficiency surfaces.

Declaration of Generative AI Use

Generative AI tools were used to support code development, debugging, figure preparation, and draft writing. All text, code, figures, and numerical results were reviewed by the authors, and the authors take full responsibility for the content of this paper. No generative AI tool was used to create or alter the underlying compressor map data.

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